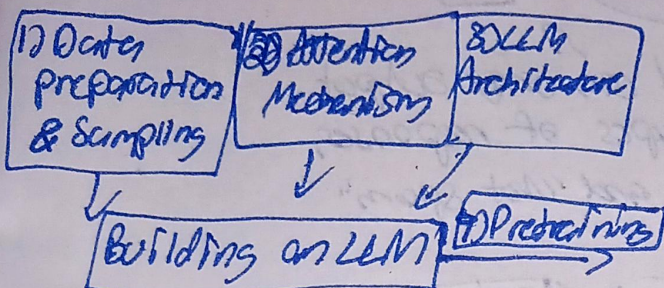
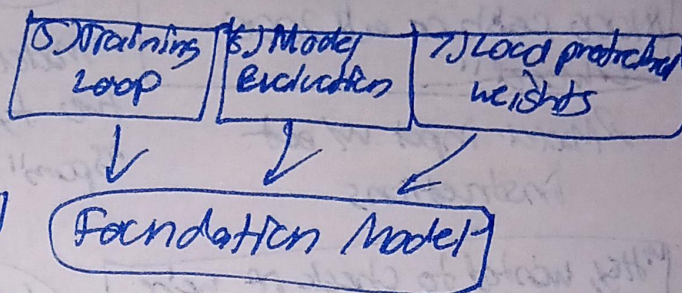


LLM Finetuning Approaches

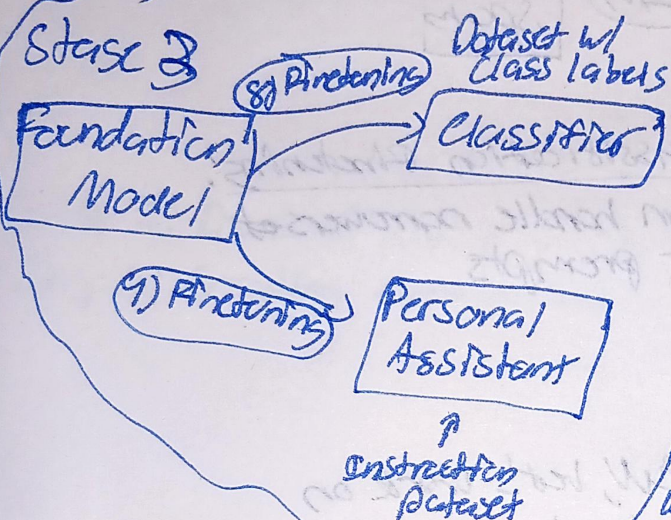
Stage 1 ✓



Stage 2 ✓



Stage 3



Finetuning: Adapting a pretrained model to a specific task by training the model on additional data.

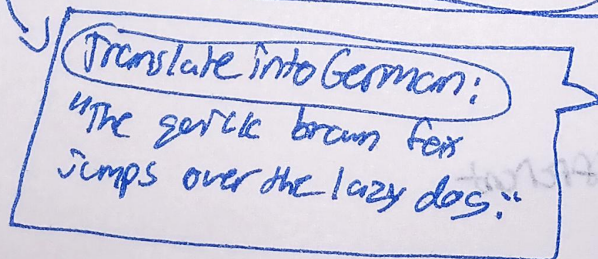
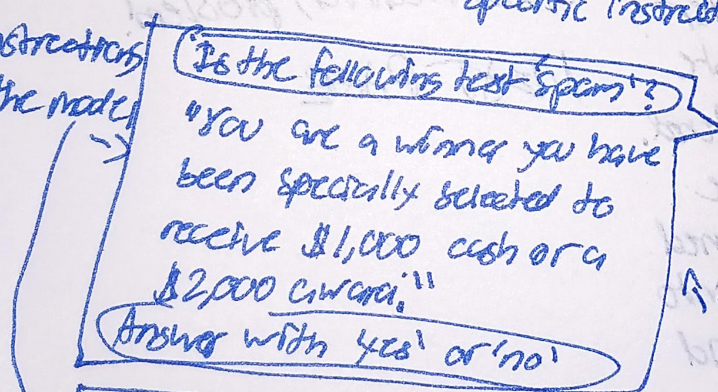
Instruction Finetuning

Classification Finetuning

Training a language model on a set of tasks using specific instructions

Model is trained to recognize a specific set of class labels, such as spam or no spam

Add instructions for the model



→ NEXT PAGE

Classification finetuning example

"You are a winner you have been selected to receive \$1000 cash or a \$2000 reward!"

Model Input w/ act instructions

LLM

spam

Model can only output two types of responses, "spam" and "not spam"

"Hey, wanted to check if we're still on for dinner tonight? Let me know!"

LLM

not spam

Instruction finetuning:

Can handle broader set of tasks

↳ Larger datasets, greater computational power needed

LoRA: Instead of finetuning all the weights that constitute the weight matrix of the pre-trained LLM, two smaller matrices that approximate this larger matrix are fine-tuned. These matrices constitute the LoRA adapter. This fine-tuned adapter is then loaded into the pre-trained model and used for inference.

↳ Low-Rank Adaptation

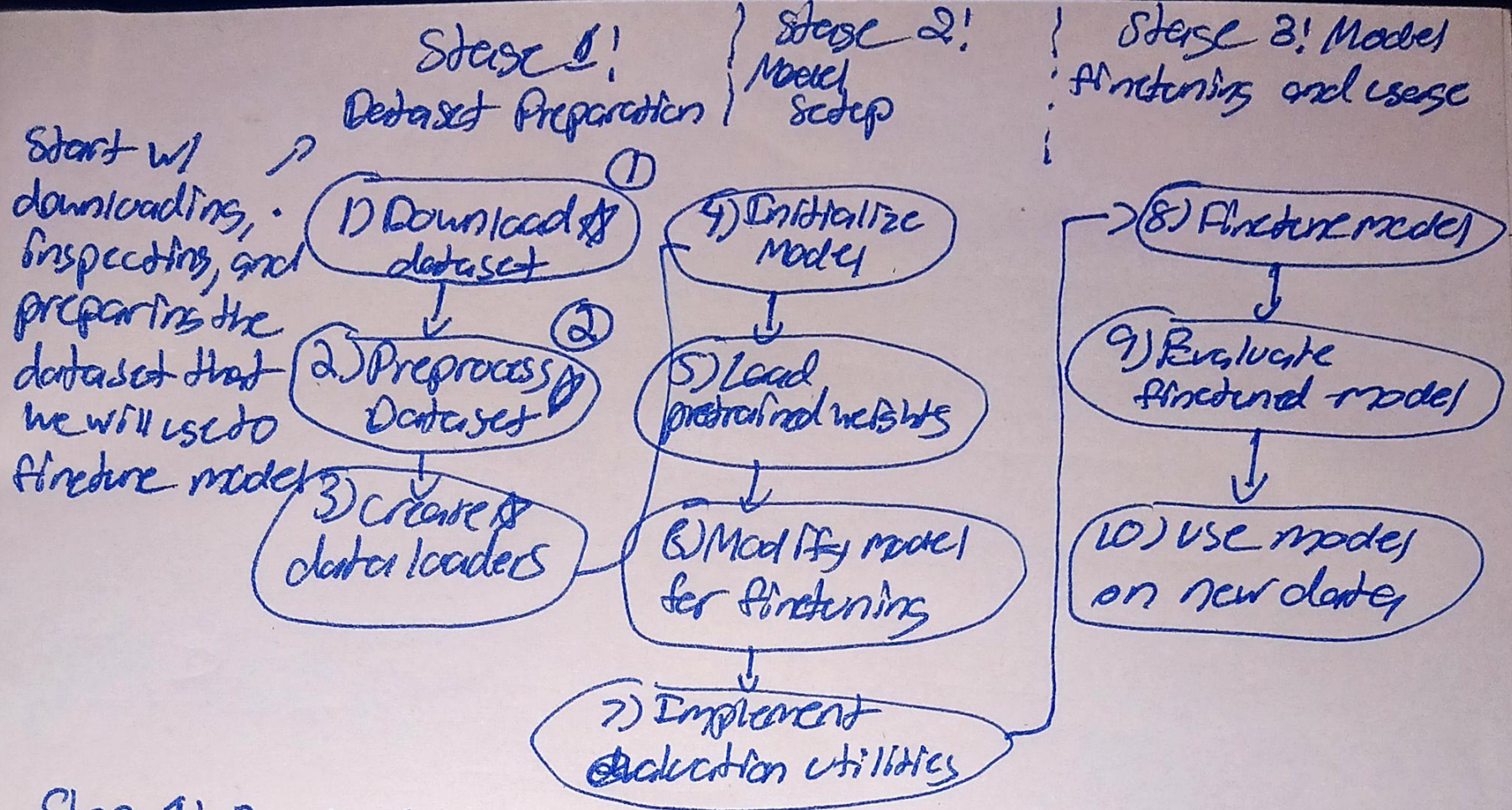
↳ LoRA → More memory-efficient iteration of LoRA

Classification finetuning:

Can handle narrower set of prompts

Now, let's work on a finetuning classification problem!

NEXT PAGE



Step 1: Download and preprocess dataset

Step 2: Creating data loaders